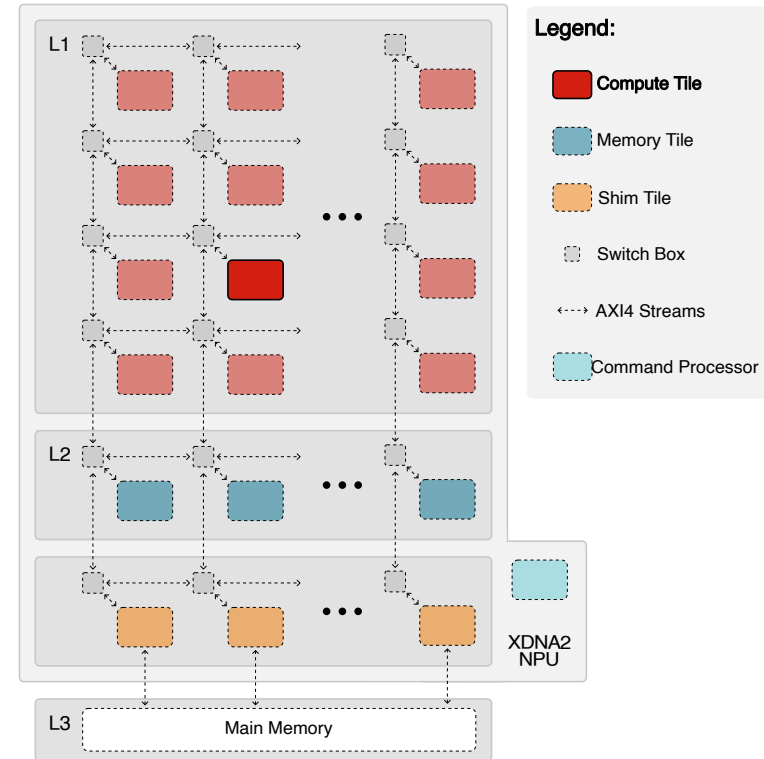




XDNA Compute Tile

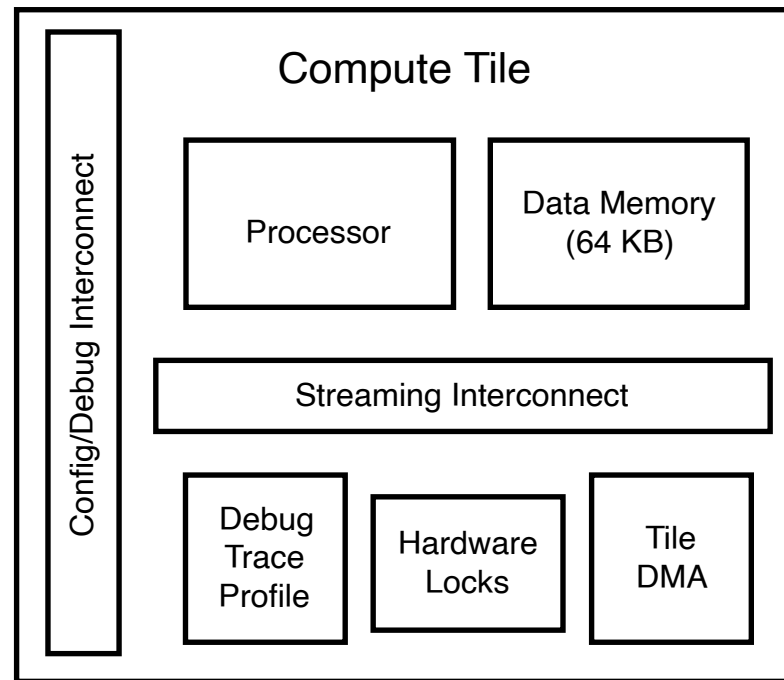
Tamino Steinert

Compute Tile



Compute Tile

- Programmable processor
- 16 KiB instruction memory
- 64 KiB data memory (L1 scratchpad memory)
- 16× 6-bit hardware locks
- Two DMA input streams and two DMA output streams

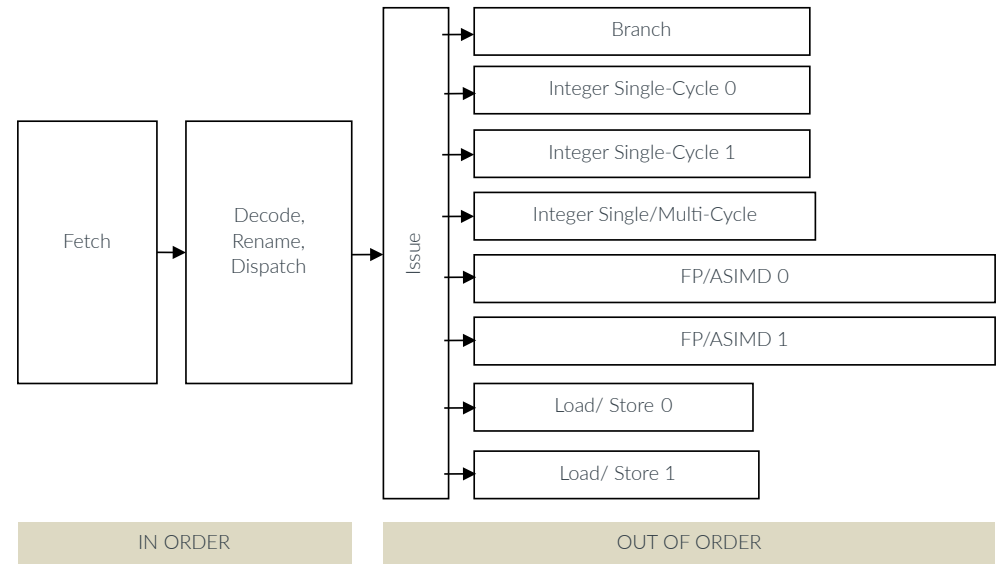


Compute tile overview.
(adapted from AMD – AIE-ML v2 Architecture Manual – AM027)

Comparison Between CPUs (e.g. Arm Cortex-A76) and XDNA Processor

Similarities Between CPUs and XDNA Processors

- Pipelined processor:
 - Units can consist of multiple stages
 - Each stage performs a different part of the operation
 - A new instruction can be issued on the unit before the previous is finished
- Multiple functional units
 - Specialized functional units, e.g. arithmetic-logic unit (ALU), load unit, store unit, control unit, ...
- Instruction set architecture (ISA)



Pipeline Overview of Arm Cortex-A76 (Raspberry Pi 5).
(by Arm – Arm Cortex-A76 Software Optimization Guide)

Instruction Scheduling

CPU – Automatic Instruction Scheduling

- Hardware tracks functional units and decides where to schedule instructions
- Instructions are automatically scheduled on idle functional units

XDNA Processors – Explicit Instruction Scheduling

- Instructions must be explicitly scheduled on functional units

Example

- Both have load units A and B

```
1 ld r0, [p0]    # issued on load unit a or b
2 ld r1, [p1]    # issued on load unit a or b
```

```
1 lda r0, [p0]   # issued on load unit a
2 ldb r1, [p1]   # issued on load unit b
```

Execution Order

CPU – Out-of-order execution

- Can reorder instructions without changing the result

XDNA Processors – In-order execution

- Always executes instructions in order

Example

```
1 ld r2, [p0]
2 mul r3, r2, #2
3 mov r1, #2
```

- Instruction 2 must wait for instruction 1 to finish
- Instruction 3 can be issued before instructions 1 and 2
- Executes instructions in order – 1, 2, 3

Instruction-level Parallelism (ILP)

CPU – Hardware ILP

- Superscalar processor – can issue multiple instructions per cycle

XDNA Processors – Software ILP

- Very long instruction word (VLIW) – instructions can have multiple operations

Example

- Both have one integer unit and two load units A and B
- Both can issue the following code in one cycle:

```
1 mul r5, r2, r2 # cycle 1 to integer unit
2 ld r0, [p0]    # cycle 1 to load unit a
3 ld r1, [p1]    # cycle 1 to load unit b
```

```
1 mul r5, r2, r2; lda r0, [p0]; ldb r1, [p1] # cycle 1
```


Data and Control Hazards

CPU – Hazard Unit

- Resolves data and control hazards automatically
- Executes an instruction when dependencies are resolved

XDNA Processors – No Hazard Unit

- Ignores data and control hazards
- Executes next instruction every cycle

Example

```
1 mov r0, #2      # writes after 1 cycle.
2 mul r0, r0, r0   # writes after 2 cycles.
3 add r1, r0, #2   # writes after 1 cycle.
4 mul r0, r1, r0   # writes after 2 cycles.
5 nop
```

- **r0** has value: **24**
- **r1** has value: **6**

- **r0** has value: **16**
- **r1** has value: **4**

Caches and Scratchpad Memory

CPU – Hardware-managed Cache

- Load/store instructions refer to main memory
- Automatic propagation of data through cache levels
- Generic replacement strategy

XDNA Processors – Software-managed Cache (Scratchpad Memory)

- Load/store instructions refer to the nearest scratchpad memory (L1)
- Explicit data movement through scratchpad levels
- Handcrafted replacement strategy

Summary of Differences

CPU's

- Automatic Instruction Scheduling
- Out-of-order execution
- Hardware ILP – superscalar processor
- Hazard unit
- Hardware-managed cache

Pros

- Easier to write code
- Better performance with bad code

XDNA Processors

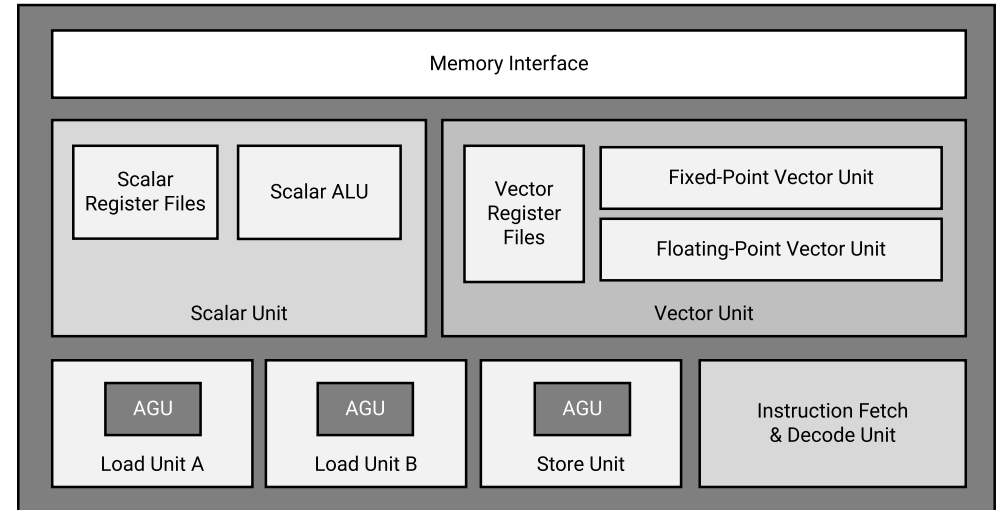
- Explicit Instruction Scheduling
- In-order execution
- Software ILP – VLIW
- No hazard unit
- Software-managed cache

Pros

- More control over hardware
- Simpler hardware design

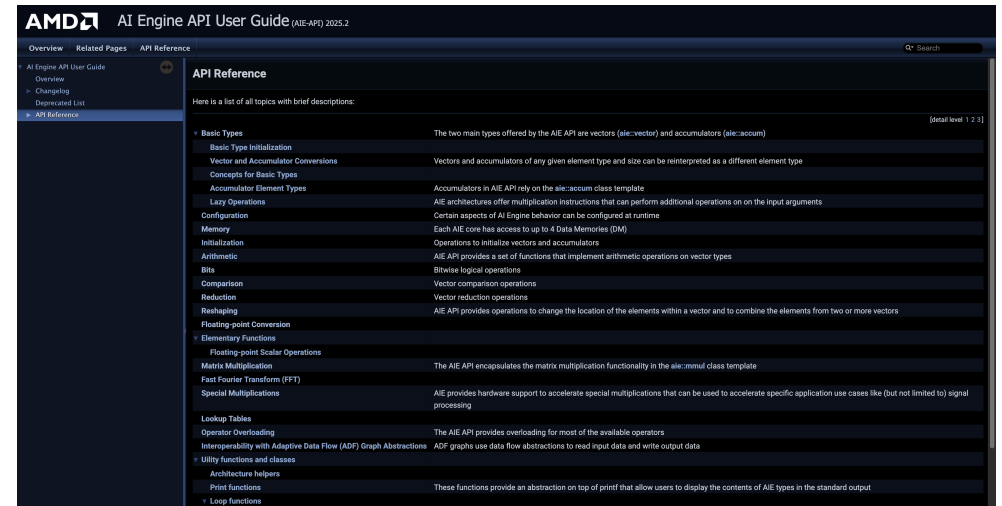
XDNA2 Processor

- Functional units:
 - 32-bit scalar unit
 - 1024 FLOPs/cycle (bfp16ebs8) vector unit
 - Two 512-bit load units with address generator units (AGUs)
 - One 512-bit store unit with AGU
 - 512-bit move unit
 - 128-bit Instruction unit
- Register files:
 - $32 \times$ 32-bit scalar registers
 - $6 \times$ 1024-bit vector registers
 - $5 \times$ 2048-bit accumulator registers

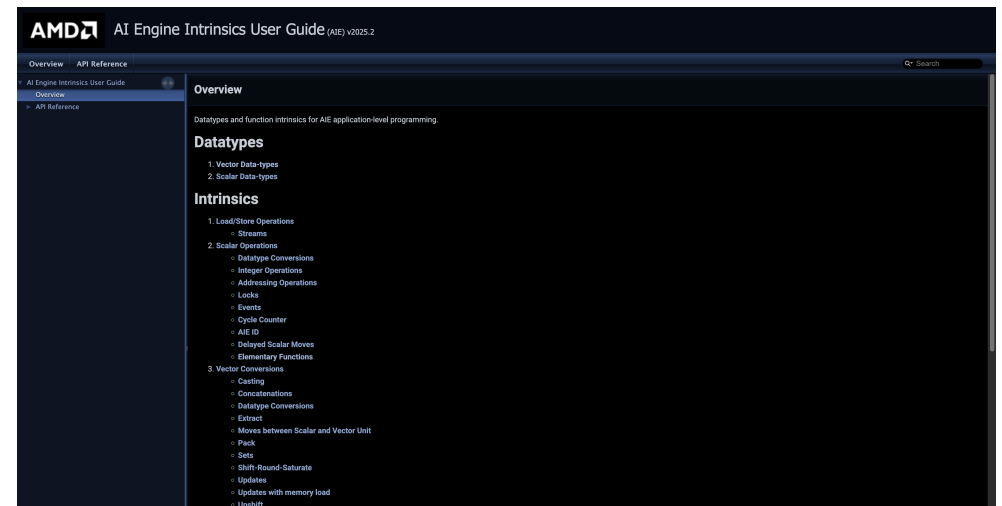


Functional units of an XDNA Processor.
(by AMD – AIE-ML v2 Architecture Manual – AM027)

XDNA ISA



AMD – AI Engine API User Guide – AIE-API



AMD – AI Engine Intrinsics User Guide – AIE